INVITED REVIEW

Sleep/Wake Disorders After Sports Concussion: Risks, Revelations, and Interventions

Madeleine M. Grigg-Damberger

Department of Neurology, University of New Mexico School of Medicine, Albuquerque, New Mexico, U.S.A.

Summary: Sleep-wake disturbances (SWDs) are among the most prevalent, persistent, and often disregarded sequelae of traumatic brain injury. Identification and treatment of SWDs in patients with traumatic brain injury is important and can complement other efforts to promote maximum functional recovery. SWDs can accentuate other consequences of traumatic brain injury, negatively affect mood, exacerbate pain, heighten irritability, and diminish cognitive abilities and the potential for recovery. The risk for sports injuries increases when athletes are sleep deprived. Sleep deprivation increases risk-taking behaviors, predisposing to injuries. SWDs are an independent risk factor for prolonged recovery after sports-related concussion. SWDs following sports-related concussion have been shown to impede recovery, rehabilitation, and return to preinjury activities.

Key Words: Sports-related concussion, Insomnia, Hypersomnia, Actigraphy, Polysomnography.

(J Clin Neurophysiol 2023;40: 417–425)

The Centers for Disease Control and Prevention estimate that between 1.6 and 3.8 million sports-related traumatic brain injuries (TBI) occur each year in the United States.[1] Approximately 80 to 90% of traumatic brain injuries related to sports are concussions.[2] A recent study combining data from emergency department visits, office visits, and a high school injury surveillance system estimated that 1.1 to 1.9 million sports-related concussion (SRC) occur annually in the US children <18 years.[3] Only 20% of SRC are related to organized team sports, and 2% to 15% of athletes participating in organized sports will suffer a concussion during one season.[4] Concussion risk varies by sport, level (high school vs. college), and sex.[5] Sports associated with highest risk for concussion among men are football, rugby, ice hockey, wrestling, and lacrosse; soccer and basketball in women.[5,6]

Concussion is a subset of mild TBI (mTBI) induced by biomechanical forces (acceleration, deceleration, and/or rotation) from a blow to the head, face, neck, or body that causes an alteration in mental status and one of more of the following symptoms: headache, nausea, vomiting, dizziness/balance problems, fatigue, difficulty sleeping, drowsiness, sensitivity to light/noise, blurred vision, memory deficits, difficulty concentrating, and disturbances in mood and sleep.[7,8] SRC can occur with or without loss of consciousness (LOC) or posttraumatic amnesia; only 9% of concussed individuals experience LOC, 23% posttraumatic amnesia.[7] LOC if it occurs in SRC typically lasts <20 minutes and posttraumatic amnesia <24 hours. Symptoms of SRC usually resolve in 80% to 90% of older adolescents and adults within 2 weeks,[9] 4 weeks in children,[10] but symptoms persist in 10% to 30%.[4,11,12] Table 1 summarizes the variety of physical, cognitive, emotional, and sleep symptoms observed with varying frequency and severity in patients with SRC.

The pathophysiology of concussion is not completely understood and based primarily on animal models extrapolated to humans.[4] The biomechanical force delivered to the brain during a concussion triggers a complex cascade of ionic, metabolic, hemodynamic, structural, and electrophysiologic changes.[13–15] Mechanical disruption of cell membranes and axonal stretching causes unregulated efflux of ions through previously regulated ion channels. This rapid depolarization provokes indiscriminate release of numerous neurotransmitters but primarily glutamate. The Na/K ATP-dependent pump tries to re-establish ionic balance which often leads to depleted energy stores. Decreased cerebral blood flow has been observed at 1 day and 1 week after SRC in adult athletes compared with controls and which recovered by 1 month.[16]

Sleep-wake disturbances (SWDs) are among the most prevalent, persistent, and until recently often disregarded sequelae of TBI.[17] Sleep problems are one of the most common complaints after SRC, affecting 30% to 70% of patients.[18–20] The aim of this review was to summarize the prognostic significance of sleep/wake complaints and SWD on prognosis of SRC, the role of clinical neurophysiology testing in evaluating them, and the value of restorative sleep before and after sports-related injuries.

SLEEP/WAKE DISORDERS AFTER SPORTS-RELATED CONCUSSION AND MILD TRAUMATIC BRAIN INJURY PREDICTS PROTRACTED RECOVERY AND GREATER FUNCTIONAL IMPAIRMENT

Poor sleep quality during the acute phase of SRC has been shown to predict lingering postconcussion symptoms, more mood disturbances, poorer community integration, and more cognitive disability 1 year after injury.[21] Sleep/wake complaints after SRC impede recovery and return to preinjury activities.[21–26]

The author has no funding or conflicts of interest to disclose.

Parts of this review article were first presented by Madeleine Grigg-Damberger MD at Joint ACNS/DGKN Symposium, The Role of Neurophysiology for Acute Sports Injuries at the ACNS meeting in Phoenix, AZ, February 12, 2017.

Address correspondence and reprint requests to Madeleine Grigg-Damberger, MD, Department of Neurology, University of New Mexico School of Medicine, MSC10 5620, 1 University of New Mexico, Albuquerque, NM 87131-0001, U.S.A.; e-mail: mgriggd@salud.unm.edu.

ISSN: 0736-0258/23/4005-0417
DOI 10.1097/WNP.0000000000000931

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

TABLE 1. Symptoms of Sports-Related Concussion

Autonomic	Cognitive	Vestibular	Ocular	Sleep and Mood
Headache	Memory difficulties	Dizziness with movement or change in position	Convergence insufficiency	Anxiety, depression, rumination
Phonophobia	Confusion and mental fogginess	Nausea and/or vomiting	Accommodation deficits	Limited social interaction, anhedonia, sadness
Photophobia	Slowed reaction times	Balance problems	Headache triggered by visual activity	Difficulty falling and/or staying asleep
Postural hypotension	Shorter attention span	Vertigo when lying down, rolling over, looking up	Difficulty reading	Needing more sleep (pleiosomnia)
Pain	Impaired concentration	Motion sensitivity		Daytime naps and irregular sleep/wake patterns

The most common sleep/wake complaints after SRC are difficulty falling and/or staying asleep (insomnia).[27] Others complain of requiring more sleep than they did before the injury (pleiosomnia).[28] Hypersomnia is reported after mild TBI but much more common in following moderate to severe TBI.

One study found that 45% of 261 children and adolescents reported trouble falling asleep ≤10 days after SRC.[29] Trouble falling asleep was associated with a high symptom burden and poorer clinical outcome. Another study of 545 adolescents attending a sports medicine clinic for SRC found those who reported sleeping <7 hours the night had significantly more postconcussive symptoms on the Post-Concussion Symptom Scale (PCSS).[30]

Two other studies found a complaint of insomnia after SRC increased the risk for persistent postconcussion syndrome 2.19-fold in a study of 1,507 National Collegiate Athletic Association (NCAA) student athletes[31] and 2.79-fold in 1,334 concussed high school athletes.[32] A 3-fold to 4-fold increased recovery time was observed among 417 adolescents who reported disturbed sleep after SRC.[26] Sleeping poorly <7 days and >3 weeks after SRC in a prospective case–control study of 997 middle and high school athletes correlated with greater migraine, cognitive, and neuropsychological symptoms and was associated with prolonged recovery times.[24] A prospective longitudinal study of 238 patients with mild TBI found those who reported "sleeping poorly" 6 months after injury were 7.7 times more likely to have residual functional impairments.[33] Figure 1 summarizes the complex relationships (often bidirectional) between sleep problems after SRC and recovery. Sleep/wake disorders following concussion predict cognitive recovery, anxiety, depression, and pain after TBI, suggesting causal role of sleep/wake disorders in subsequent recovery.

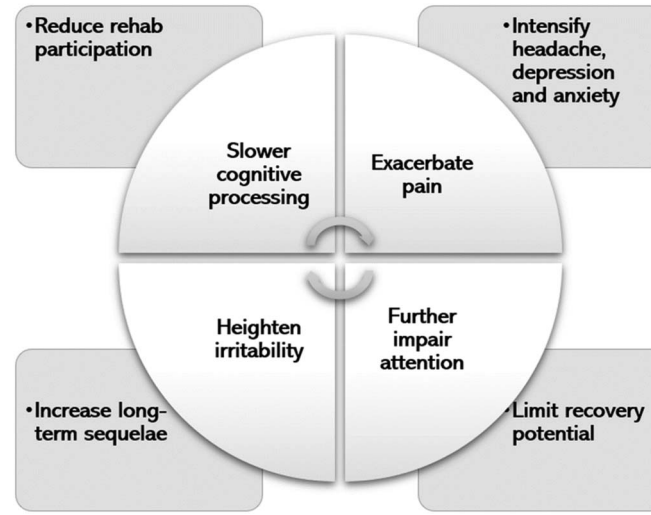

FIG. 1. Complex relationships (often bidirectional) between sleep problems after sports-related concussion and recovery. Sleep/wake disorders after concussion predict cognitive recovery, anxiety, depression, and pain after TBI, suggesting causal role of sleep/wake disorders in subsequent recovery.

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

COMPREHENSIVE IN-LABORATORY POLYSOMNOGRAPHY AFTER MILD TRAUMATIC BRAIN INJURY OFTEN UNREVEALING

The few studies recording polysomnography after mild TBI usually do not find objective alterations in gross sleep architecture. A 2018 meta-analysis of 14 studies comparing 305 patients with chronic TBI still reporting sleep/wake complaints >6 months after injury compared with 288 age-matched controls without a history of TBI or sleep/wake complaints found that (1) moderate to severe TBI was associated with an increase in non-rapid eye movement sleep (NREM) 3 sleep time, reduced NREM 2, and (2) sleep efficiency, but mild TBI was not associated with any significant alteration of sleep architecture.[34]

A recent cross-sectional study recorded polysomnogram (PSG) in 40 adults (mean age 47 ± 11 years) diagnosed with mild TBI, comparing their results with population norms.[35] In patients with mTBI, the mean duration of nocturnal wakefulness was higher and consolidated NREM 2 and REM sleep was lower. However, another study recorded overnight PSG in 21 athletes, 10 with a mean of four concussions including one in the past year and 11 with no history of concussion.[36] They recorded two consecutive nights of PSG (to accommodate for so-called first night effects on sleep architecture when sleeping in the foreign environment of the sleep laboratory for the first time). Although the concussed athletes reported more sleep/wake complaints, no between-group differences were found in sleep macroarchitecture or microarchitecture (sleep latency, sleep efficiency, percentages of different sleep stages, numbers of arousals, awakenings, and wake after sleep onset).

QUANTITATIVE EEG POWER STUDIES AFTER SPORTS-RELATED CONCUSSION AND MILD TRAUMATIC BRAIN INJURY

A few studies have used the power spectral analysis to evaluate quantitative EEG (QEEG) power during sleep to identify more subtle changes in sleep architecture.[36–40] These performed in subjects after mild TBI (not specifically SRC) have shown conflicting results. One study showed decreased delta power (a surrogate for NREM 3 sleep) and increased alpha and beta power (surrogates for awakening and arousal) during NREM sleep in seven subjects one week after mild TBI compared with seven controls.[37] Increases in occipital beta power during sleep were observed in 34 adults with mild TBI a mean of 11 weeks after injury compared with 29 healthy controls.[38] Significantly more delta and less beta power only when awake was observed on PSG in 10 athletes who had repeated SRC in the past year when compared with 11 nonconcussed athletes.[36]

Perhaps most interesting is the study which correlated increased beta and gamma power in sleep in mild TBI associated with pain and sleep complaints. Khoury et al.[39] recorded overnight PSG in 24 patients with mild TBI complaining of sleep/wake problems (some also complaining of pain) and compared them with 18 healthy controls without sleep or pain complaints. They found increased beta and gamma power in the frontal, central, and occipital derivations in REM sleep, and beta power in NREM sleep was more likely to occur in patients with mild TBI who reported pain and sleep problems. Increased beta and gamma power in sleep after mild TBI may be a biomarker for unrefreshing sleep and the sleep-disrupting effects of pain after mild TBI.

WRIST ACTIGRAPHY MOST USEFUL AS OBJECTIVE PROXY TO STUDY SLEEP AFTER MILD TRAUMATIC BRAIN INJURY

Actigraphy is often used as an objective proxy for sleep after mild TBI including SRC. Actigraphy identifies sleep and wakefulness based on the tenet that body movements when awake are frequent and large and absent or small during sleep.[41,42] Actigraphs resemble wrist watches and are most often worn on the nondominant wrist. Actigraphs use analog linear accelerometers to detect body movement, digitalize this data, tally it typically in 1 minute epoch intervals, and store it into internal memory. The data from the continuous actigraphic recordings are downloaded to a specially designed computer program which writes out a graphic representation of activity and inactivity over extended periods of time to estimate sleep/wake patterns.

Actigraphy provides information about the day-to-day timing, duration, and continuity of sleep in the natural sleep environment over long periods (typically 1–2 weeks, sometimes months). Patients must keep daily sleep logs detailing their bedtimes and potential artifact-producing activities to reduce data misinterpretation. Data from these provide estimates of sleep latency, total sleep time (TST), wake time, wake after sleep onset, and pattern of sleep/wakefulness over days of recording. The results of actigraphy are greatly improved by requiring patients to keep a sleep log documenting bed times and when the device is off.

Actigraphy has been used to examine the relationships between insomnia, pain, increased sleep need, and mild TBI. A prospective study recorded actigraphy for two 5-day sessions (day of injury and 30 days later) coupled with sleep diaries in 17 young adults (7 with acute SRC and 10 controls).[43] They found TST, and coefficient of variation for TST (greater nighttime sleep duration variability) was significantly greater in the postconcussive group. The authors believed these findings suggested that individuals with a concussion may require and obtain more sleep shortly after injury and subsequently have a shorter duration of sleep at 1 month after injury.

Another study recorded actigraphy in 20 athletes (ages 12–19 years) correlating sleep/wake patterns and quantities with symptom, neurocognitive, and vestibular/oculomotor scores.[44] They explored the relationships between excessive physical activity on actigraphy and SRC recovery between visit 1 (<72 hours after injury) and visit 2 (6–18 days after injury). Excessive physical activity and poor sleep in the first week after SRC tracked using actigraphy were associated with worse reaction times, vestibular/oculomotor scores, and outcomes at the second visit (6–18 days after injury). The authors concluded that excess

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

physical activity and poor sleep the first week after injury may be associated with worse outcomes at follow-up in the subacute stage of recovery.

A third study recorded 7 ± 2 consecutive days of actigraphy in 56 individuals 1 month after mild TBI, 24 at 1 year after mild TBI, and 20 controls to examine the relationships between sleep/wake activity, insomnia, and pain after mild TBI.[45] They found that patients with mild TBI who reported moderate to severe pain were more likely to require more than 8 hours of sleep per day (37 vs. 11%) and take three or more naps per week (42 vs. 22%). Pain and an increased sleep need (sleeping more than 8 hours and/or frequent naps) were found in 29% of patients with mild TBI 1 month after injury.

Significantly shorter sleep duration, poorer sleep efficiency, and more wake after sleep onset were observed among 50 adolescents with mild TBI 3 to 12 months after injury and compared them with 50 healthy controls.[46] For both groups of adolescents, poorer self-reported sleep quality was predicted by greater depressive symptoms.

Actigraphy in mild TBI has been shown to be useful for identifying circadian rhythm disorders masquerading as insomnia. One study found circadian rhythm disorders in 15 (36%) of 42 patients with mild TBI who reported insomnia.[47] Eight displayed a delayed sleep-wake phase disorder and seven irregular sleep-wake phase disorder. On a rare occasion, a non-24 hour sleep/wake disorder has been observed after TBI, best confirmed by ≥4 to 6 weeks of actigraphy and sleep diaries.[48] Figure 2 is an actigraph recorded for 1 week in an adolescent with SRC reporting insomnia. The actigraph instead shows delayed sleep-wake phase disorder.

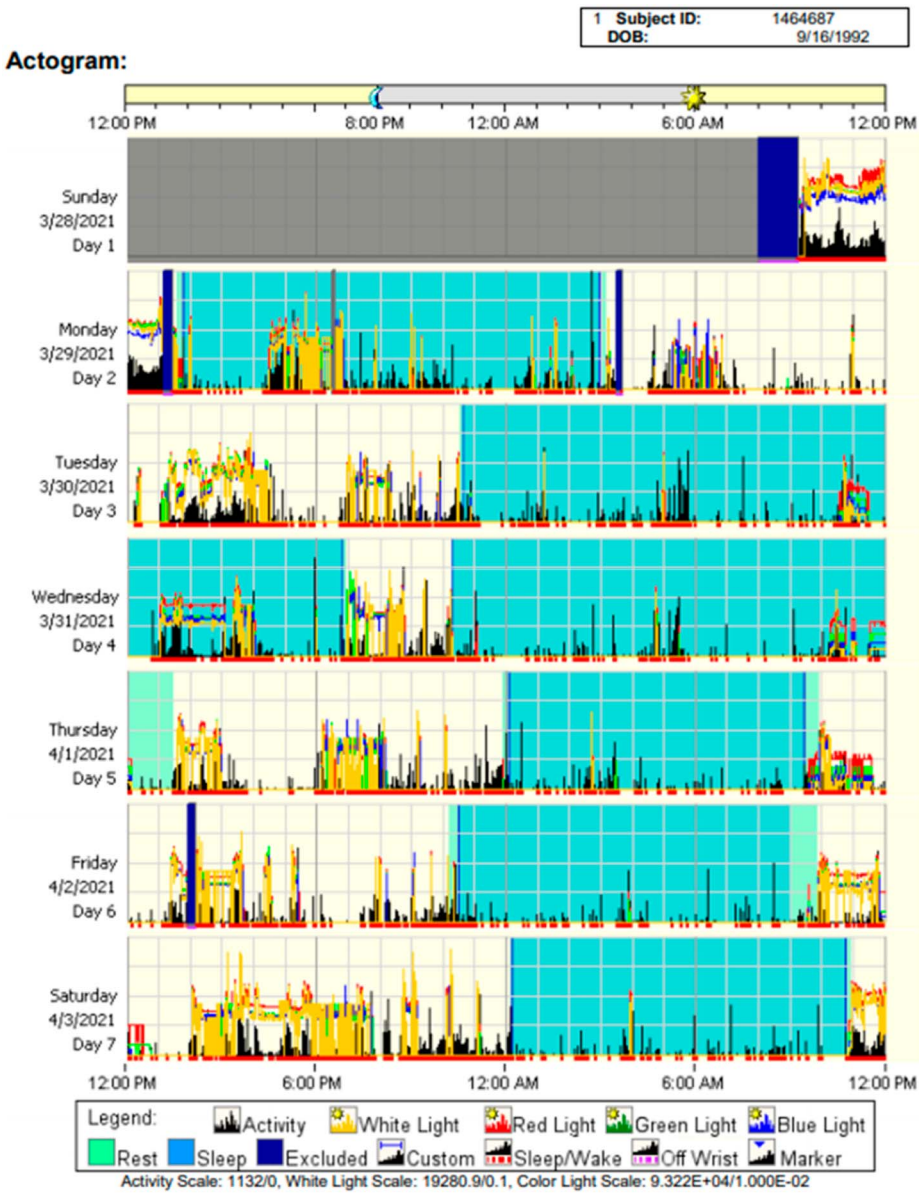

FIG. 2. One-week actigraph showing delayed sleep-wake phase disorder in an adolescent reporting insomnia after sports-related concussion.

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

PATHOPHYSIOLOGY OF SLEEP DISORDERS IN TRAUMATIC BRAIN INJURY

What predisposes individuals with mild TBI to sleep/wake disturbances? Damage to brain structures important in sleep/wake regulation including the upper brainstem, hypothalamic–pituitary axis, medial temporal lobe, basal ganglia, and white matter tracts are posited.[11] Diffuse axonal injury from concussion preferentially affects midline structures which are involved in sleep-wake regulation, i.e., deep gray matter, dorsolateral pons, and midbrain.[49] Longer tentorial length and shorter tentorial angle on brain MRI were associated with SWDs in 64 consecutive patients with mild TBI.[50] Smaller tentorial angles and SWD in patients with mild TBI inversely correlated with length of time to recover.

Another study using diffusion weighted imaging found reduced white matter integrity correlated with sleep/wake complaints in 34 individuals (mean age 24 ± 7 years) within 1 year of a mild TBI compared with 18 healthy controls.[51] They found that fractional anisotropy demonstrated a negative correlation with sleep quality scores (higher fractional anisotropy was associated with better sleep quality). Higher radial diffusivity was associated with worse sleep quality. These correlations were observed bilaterally in the anterior and posterior limbs of the internal capsule, anterior and superior corona radiata, fornix, and superior fronto-occipital fasciculi.

PREINJURY SLEEP LOSS PREDISPOSES ATHLETES TO SPORTS-RELATED INJURIES

Insufficient sleep quality and quantity may predispose athletes to sports-related injuries. A study of 160 adolescent athletes showed those who slept on average <8 hours per night were 1.7 times more likely to have a sports-related injury than those who slept ≥8 hours.[52] A questionnaire-based study of 360 athletes (ages 6–18 years) showed excessive play/training and sleeping ≤6 hours the night before increased the risk for injury.[53] Decreased sleep duration coupled with increased training load increased the risk for injury in another study of 493 student athletes.[54] Whereas sleeping >8 hours on school nights among 340 elite adolescent athletes reduced their odds of injury 61% in another study.[55]

Athletes who were not well-slept before the injury were more likely to have more symptoms and greater cognitive impairment after injury. Among 348 adolescent athletes, those who endorsed preinjury sleep difficulties before SRC showed (1) higher total and sleep-related symptoms across postinjury time intervals, (2) poorer verbal memory performance on day 2 after injury, and (3) slower reaction times on 5 to 7 days and 10 to 14 days.[56]

A 2020 study evaluated the impact of sleep on the relationship between soccer heading exposure and neuropsychological function in 50 varsity and intramural college-age soccer players.[57] With reduced sleep duration, a high level of recent heading exposure was related to worse sustained attention.

How could insufficient sleep predispose to sports-related injuries? Recent research reports that sleep deprivation predisposes adolescents and young adults to risk behaviors, diminished attentional and behavioral control, poor emotion regulation, reward-seeking, and love of gambling and winning.[58–61] A 2018 meta-analysis of 24 studies and more than 50,000 subjects found insufficient sleep in adolescents was associated with 1.4-fold greater odds of risk-taking in adolescents.[61] A seminal study by Telzer et al.[60] using functional brain MRI in 46 adolescents performing a cognitive control and risk-taking task showed increased risk tasking was paralleled by less recruitment of the dorsolateral prefrontal cortex during cognitive control, greater insular activation during reward processing, and reduced functional coupling between dorsolateral prefrontal cortex and affective regions in the insula and ventral striatum during reward processing.

CONSUMER WEARABLES USED TO MONITOR SLEEP IN ATHLETES

Can the data from a consumer wearable provide meaningful information about sleep in athletes? A recent study measured TST in 12 well-trained soccer players (mean age 18 years) using a consumer wearable.[62] The results were compared with some simultaneous nighttime PSG recordings. The device automatically detected 24 (80%) of 30 nighttime sleep periods but only 6 (30%) of 20 of the 1 to 2 hours of daytime naps. Compared with PSG, the device overestimated TST by an average of 52 ± 152 minutes for nighttime sleep periods and by 4 ± 8 minutes for daytime naps. If athletes use wearable sleep devices to measure TST, it is best not to depend on the automatic sleep tracking data alone. Instead, the start/end times of sleep periods should be manually checked each day and adjustments made where required. Using this methodology reduced the average difference in TST between the device and PSG to 7.0 ± 22 minutes.

ROLES OF REST AND SLEEP IN RECOVERY FROM SPORTS CONCUSSION

Prescribing cognitive and physical rest has been a mainstay of treatment for SRC for several decades, despite insufficient evidence to support this approach.[4,63] Recent data in aerobically trained animals given early access to exercise showed better outcomes compared with no or delayed exercise or to social isolation.[64] Total rest ("the dark room" or "cocoon therapy") is no longer recommended.[11,65] Consensus guidelines endorse 24 to 48 hours of symptom-limited cognitive and physical rest followed by a gradual increase in activity, staying below symptom exacerbation thresholds.[11,65] Emerging data suggest that a symptom-limited activity, including activities of daily living and noncontact aerobic exercise, may begin after a 24 to 48 hours of cognitive and physical relative rest.[66–68] Figure 3 shows a graduated return to sport strategy.[66,67] A graduated return to learning is also worthy of consideration including extension of assignment deadlines, rest periods during the school day, postponement or staggering of test, and accommodation for light or noise sensitivity.[68]

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

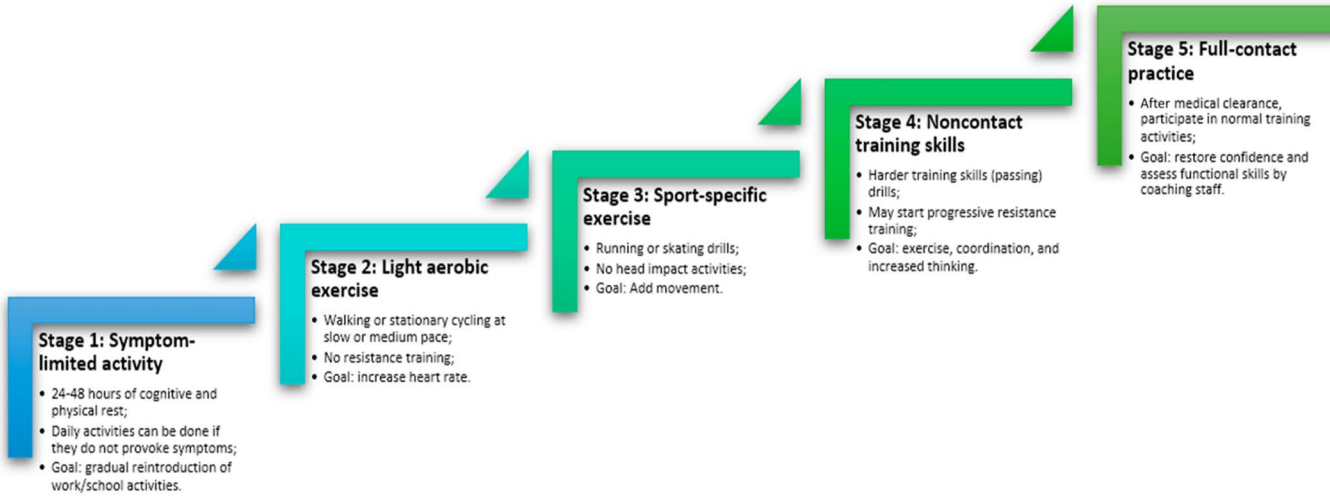

FIG. 3. Graduated return to sports strategy.

IDENTIFYING AND TREATING SLEEP/WAKE DISORDERS IMPORTANT FOR RECOVERY

Recently published clinical practice guidelines emphasize the importance of screening for symptoms of sleep/wake complaints and SWDs after SRC and mild TBI.[69–71] Identifying these warrants screening for predisposing and/or comorbid medical, psychiatric conditions and medication effects. If sleep/wake complaints persist, referral to a sleep specialist (preferably with experience in managing SWDs after TBI) is recommended. Comprehensive in-laboratory PSG is worthy of consideration if sleep/wake complaints persist and/or sleep-related breathing disorders, nocturnal seizures, periodic limb movements, or narcolepsy suspected.

Treating SWDs may positively affect other persistent symptoms (mood, anxiety, pain, fatigue, and cognitive problems) and shorter recovery time. The first treatment goal was to establish a more routine sleep pattern. Insomnia is first best treated with stimulus control therapy, sleep restriction strategies, and cognitive behavior therapy for insomnia. Medications to induce or sustain sleep are best avoided in TBI. If needed, best to use medications which will not produce dependency and have minimal adverse effects for patients with mTBI.

A 2021 study examined the value of health-promoting behaviors (such as sleep duration, diet quality, and exercise frequency) on self-reported measures of brain health (cognitive function, symptoms of anxiety and depression, and emotional/behavior control) among 1,784 National Football League athletes who reported SRC.[72] They found sleeping <6 hours per night was associated with more depressive symptoms, anxiety, and emotional/behavior dyscontrol. Obtaining adequate sleep may be a therapeutic or even preventative intervention for athletes who have suffered SRC.

A 2021 double-blind placebo-controlled trial evaluated the effect of exposure to 30 minutes of morning blue light on brain structure, functional connectivity, and daytime sleepiness in 62 individuals after mTBI.[73] Exposure to blue light was associated with moderate to large increases in both gray matter volume and functional connectivity in areas and networks associated with sleep regulation and daytime cognitive function, alertness, and attention. Individuals receiving blue light had greater gray matter volume than those receiving amber (placebo) light in the right thalamus and bilateral prefrontal and orbitofrontal cortices. Subjects who reported improved daytime sleepiness had (1) greater increases in gray matter volume in the same regions, (2) increased functional connectivity between the thalamus and both prefrontal and orbitofrontal cortices, (3) increased functional connectivity between attention and cognitive control networks, and (4) decreased connectivity between visual, motor, and attention networks.

A 2021 study analyzed the efficacy of 3 and 10 mg of oral melatonin for sleep disturbance in 72 children with persistent postconcussive symptoms and sleep-related problems from a randomized controlled trial.[74] Actigraphy was recorded in 64 children. Sleep duration increased in the melatonin 3 mg (43 minutes) and melatonin 10 mg groups (55 minutes) compared with placebo. Sleep efficiency significantly increased with the 10 mg dose. Sleep-related problems lessened.

EXPERIMENTAL SLEEP MODULATION ALLEVIATES AXONAL DAMAGE AND COGNITIVE DECLINE AFTER RODENT TRAUMATIC BRAIN INJURY

An interesting study by Morawska et al.[75] used a rat model of closed skull TBI that reproduces human TBI consequences. They assessed cognition with an objective recognition test and assessed diffuse axonal injury staining for an amyloid precursor protein. They found enhancing NREM 3 sleep after TBI markedly reduced diffuse axonal damage in the cortex and hippocampus and lessened memory impairment 2 weeks after injury. Enhancing NREM 3 sleep acutely after TBI may have beneficial disease-modifying effects after acute TBI, perhaps by NREM 3 preferential cleaning of neurotoxic proteins and other waste products.

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

SLEEP AND SLEEP DISORDERS IN FORMER ATHLETES WITH CHRONIC TRAUMATIC ENCEPHALOPATHY

Repeated head impacts particularly in collegial or professional athletes increase their risk for later developing the neurodegenerative tauopathy, chronic traumatic encephalopathy (CTE).[76] CTE is characterized by memory loss, behavioral problems, depression, dementia, and potentially premature death (including suicide).[77] Neuropathological findings are patchy perivascular accumulations of hyperphosphorylated tau protein in the depths of cerebral sulci.[78] Repeated head impacts are believed to trigger neuroinflammation, neurovascular dysfunction, neuronal abnormalities, and axonopathy leading to CTE (and other dementias including Lewy body disease).[76,78] Brain MRI shows diffuse atrophy with abnormal axonal findings using diffusion tensor imaging.[79] Increased deposition of tau is seen in the brain PET scans in individuals with CTE compared with controls.[80,81]

Sleep/wake complaints including insomnia and obstructive sleep apnea are common among them and often precede onset of the disease.[82] Whether identifying and treating insomnia and obstructive sleep apnea in them can reduce the risk for later developing CTE and other dementias (such as Lewy Body Disease) is not yet known.[83] Of note, a 2020 study found probable REM behavior disorder in 32% of 247 men with CTE; more likely in those with more years of exposure to repeated head impacts especially those more likely to also develop Lewy body disease.[84]

IN CLOSING

Sleep-wake disturbances are an independent risk factor for prolonged recovery after SRC. Sleep-wake disturbances after SRC have been shown to impede recovery, rehabilitation, and return to preinjury activities. Identification and treatment of sleep disorders in patients with SRC is important and can complement other efforts to promote maximum functional recovery. SWDs can accentuate other consequences of SRC, negatively affect mood, exacerbate pain, heighten irritability, and diminish cognitive abilities and the potential for recovery. The risk for sports injuries increases when athletes are sleep-deprived. Sleep deprivation increases risk-taking behaviors, predisposing to injuries.

REFERENCES

1. Daneshvar DH, Nowinski CJ, McKee AC, Cantu RC. The epidemiology of sport-related concussion. Clin Sports Medvii 2011;30:1–9.
2. Levin HS, Diaz-Arrastia RR, Jacobsen SJ, Ilstrup DM, Garrity JA, Gorman CA. Diagnosis, prognosis, and clinical management of mild traumatic brain injury. Lancet Neurol 2015;14:506–517.
3. Bryan MA, Rowhani-Rahbar A, Comstock RD, Rivara F; Sports Concussion Research SeattleC. Sports- and recreation-related concussions in US youth. Pediatrics 2016;138:e20154635.
4. Harmon KG, Clugston JR, Dec K, et al. American medical society for sports medicine position statement on concussion in sport. Clin J Sport Med 2019;29:87–100.
5. Pierpoint LA, Collins C. Epidemiology of sport-related concussion. Clin Sports Med 2021;40:1–18.
6. Giza CC, Kutcher JS, Ashwal S, et al. Summary of evidence-based guideline update: evaluation and management of concussion in sports: report of the Guideline Development Subcommittee of the American Academy of Neurology. Neurology 2013;80:2250–2257.
7. McCrea M, Guskiewicz KM, Marshall SW, et al. Acute effects and recovery time following concussion in collegiate football players: the NCAA Concussion Study. Jama 2003;290:2556–2563.
8. Register-Mihalik JK, Kay MC. The current state of sports concussion. Neurol Clin 2017;35:387–402.
9. McCrea M, Guskiewicz K, Randolph C, et al. Incidence, clinical course, and predictors of prolonged recovery time following sport-related concussion in high school and college athletes. J Int Neuropsychol Soc 2013;19:22–33.
10. Zemek R, Barrowman N, Freedman SB, et al. Clinical risk score for persistent postconcussion symptoms among children with acute concussion in the. Jama 2016;315:1014–1025.
11. McCrory P, Meeuwisse W, Dvořák J, et al. Consensus statement on concussion in sport-the 5th international conference on concussion in sport held in Berlin, October 2016. Br J Sports Med 2017;51:838–847.
12. Gupta A, Summerville G, Senter C. Treatment of acute sports-related concussion. Curr Rev Musculoskelet Med 2019;12:117–123.
13. Barkhoudarian G, Hovda DA, Giza CC. The molecular pathophysiology of concussive brain injury—an update. Phys Med Rehabil Clin N Am 2016;27:373–393.
14. Karton C, Hoshizaki TB. Biomechanics of sport-related neurological injury. Clin Sports Med 2021;40:19–38.
15. Howell DR, Southard J. The molecular pathophysiology of concussion. Clin Sports Med 2021;40:39–51.
16. Meier TB, Bellgowan PS, Singh R, Kuplicki R, Polanski DW, Mayer AR. Recovery of cerebral blood flow following sports-related concussion. JAMA Neurol 2015;72:530–538.
17. Duclos C, Dumont M, Wiseman-Hakes C, et al. Sleep and wake disturbances following traumatic brain injury. Pathol Biol (Paris) 2014;62:252–261.
18. Kontos AP, Sufrinko A, Sandel N, Emami K, Collins MW. Sport-related concussion clinical profiles: clinical characteristics, targeted treatments, and preliminary evidence. Curr Sports Med Rep 2019;18:82–92.
19. Singh K, Morse AM, Tkachenko N, Kothare SV. Sleep disorders associated with traumatic brain injury-A review. Pediatr Neurol 2016;60:30–36.
20. Tkachenko N, Singh K, Hasanaj L, Serrano L, Kothare SV. Sleep disorders associated with mild traumatic brain injury using sport concussion assessment tool 3. Pediatr Neurol 2016;57:46.E1–50.E1.
21. Mathias JL, Alvaro PK. Prevalence of sleep disturbances, disorders, and problems following traumatic brain injury: a meta-analysis. Sleep Med 2012;13:898–905.
22. Lau BC, Collins MW, Lovell MR. Sensitivity and specificity of subacute computerized neurocognitive testing and symptom evaluation in predicting outcomes after sports-related concussion. Am J Sports Med 2011;39:1209–1216.
23. Lucke-Wold BP, Smith KE, Nguyen L, et al. Sleep disruption and the sequelae associated with traumatic brain injury. Neurosci Biobehav Rev 2015;55:68–77.
24. Murdaugh DL, Ono KE, Reisner A, Burns TG. Assessment of sleep quantity and sleep disturbances during recovery from sports-related concussion in youth athletes. Arch Phys Med Rehabil 2018;99:960–966.
25. Guty E, Arnett P. Post-concussion symptom factors and neuropsychological outcomes in collegiate athletes. J Int Neuropsychol Soc 2018;24:684–692.
26. Bramley H, Henson A, Lewis MM, Kong L, Stetter C, Silvis M. Sleep disturbance following concussion is a risk factor for a prolonged recovery. Clin Pediatr 2017;56:1280–1285.
27. Baumann CR. Sleep and traumatic brain injury. Sleep Med Clin 2016;11:19–23.
28. Sommerauer M, Valko PO, Werth E, Baumann CR. Excessive sleep need following traumatic brain injury: a case-control study of 36 patients. J Sleep Res 2013;22:634–639.

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

29. Howell DR, Oldham JR, Brilliant AN, Meehan WP III. Trouble falling asleep after concussion is associated with higher symptom burden among children and adolescents. J Child Neurol 2019;34:256–261.
30. Howell DR, Berkstresser B, Wang F, et al. Self-reported sleep duration affects tandem gait, but not steady-state gait outcomes among healthy collegiate athletes. Gait Posture 2018;62:291–296.
31. Zuckerman SL, Yengo-Kahn AM, Buckley TA, Solomon GS, Sills AK, Kerr ZY. Predictors of postconcussion syndrome in collegiate student-athletes. Neurosurg Focus 2016;40:E13.
32. Kerr ZY, Zuckerman SL, Wasserman EB, et al. Factors associated with post-concussion syndrome in high school student-athletes. J Sci Med Sport 2018;21:447–452.
33. Kalmbach DA, Conroy DA, Falk H, et al. Poor sleep is linked to impeded recovery from traumatic brain injury. Sleep 2018;41:zsy147.
34. Mantua J, Grillakis A, Mahfouz SH, et al. A systematic review and meta-analysis of sleep architecture and chronic traumatic brain injury. Sleep Med Rev 2018;41:61–77.
35. Mollayeva T, Colantonio A, Cassidy JD, Vernich L, Moineddin R, Shapiro CM. Sleep stage distribution in persons with mild traumatic brain injury: a polysomnographic study according to American Academy of Sleep Medicine standards. Sleep Med 2017;34:179–192.
36. Gosselin N, Lassonde M, Petit D, et al. Sleep following sport-related concussions. Sleep Med 2009;10:35–46.
37. Rao V, Bergey A, Hill H, Efron D, McCann U. Sleep disturbance after mild traumatic brain injury: indicator of injury?. J Neuropsychiatry Clin Neurosci 2011;23:201–205.
38. Arbour C, Khoury S, Lavigne GJ, et al. Are NREM sleep characteristics associated to subjective sleep complaints after mild traumatic brain injury? Sleep Med 2015;16:534–539.
39. Khoury S, Chouchou F, Amzica F, et al. Rapid EEG activity during sleep dominates in mild traumatic brain injury patients with acute pain. J Neurotrauma 2013;30:633–641.
40. Kerasidis H, Simmons J. Quantitative EEG analysis in clinical practice: concussion injury. Clin EEG Neurosci 2021;52:114–118.
41. Ancoli-Israel S, Cole R, Alessi C, Chambers M, Moorcroft W, Pollak CP. The role of actigraphy in the study of sleep and circadian rhythms. Sleep 2003;26:342–392.
42. Pollak CP, Tryon WW, Nagaraja H, Dzwonczyk R. How accurately does wrist actigraphy identify the states of sleep and wakefulness? Sleep 2001;24:957–965.
43. Raikes AC, Schaefer SY. Sleep quantity and quality during acute concussion: a pilot study. Sleep 2016;38:2141–2147.
44. Sufrinko AM, Howie EK, Elbin RJ, Collins MW, Kontos AP. A preliminary investigation of accelerometer-derived sleep and physical activity following sport-related concussion. J Head Trauma Rehabil 2018;33:E64–E74.
45. Suzuki Y, Khoury S, El-Khatib H, et al. Individuals with pain need more sleep in the early stage of mild traumatic brain injury. Sleep Med 2017;33:36–42.
46. Tham SW, Fales J, Palermo TM. Subjective and objective assessment of sleep in adolescents with mild traumatic brain injury. J Neurotrauma 2015;32:847–852.
47. Ayalon L, Borodkin K, Dishon L, Kanety H, Dagan Y. Circadian rhythm sleep disorders following mild traumatic brain injury. Neurology 2007;68:1136–1140.
48. Boivin DB, Caliyurt O, James FO, Chalk C. Association between delayed sleep phase and hypernyctohemeral syndromes: a case study. Sleep 2004;27:417–421.
49. Jaffee MS, Winter WC, Jones CC, Ling G. Sleep disturbances in athletic concussion. Brain Inj 2015;29:221–227.
50. Yaeger K, Alhilali L, Fakhran S. Evaluation of tentorial length and angle in sleep-wake disturbances after mild traumatic brain injury. AJR Am J Roentgenol 2014;202:614–618.
51. Raikes AC, Bajaj S, Dailey NS, et al. Diffusion tensor imaging (DTI) correlates of self-reported sleep quality and depression following mild traumatic brain injury. Front Neurol 2018;9:468.
52. Milewski MD, Skaggs DL, Bishop GA, et al. Chronic lack of sleep is associated with increased sports injuries in adolescent athletes. J Pediatr Orthop 2014;34:129–133.
53. Luke A, Lazaro RM, Bergeron MF, et al. Sports-related injuries in youth athletes: is overscheduling a risk factor? Clin J Sport Med 2011;21:307–314.
54. von Rosen P, Frohm A, Kottorp A, Friden C, Heijne A. Multiple factors explain injury risk in adolescent elite athletes: applying a biopsychosocial perspective. Scand J Med Sci Sports 2017;27:2059–2069.
55. von Rosen P, Frohm A, Kottorp A, Friden C, Heijne A. Too little sleep and an unhealthy diet could increase the risk of sustaining a new injury in adolescent elite athletes. Scand J Med Sci Sports 2017;27:1364–1371.
56. Sufrinko A, Pearce K, Elbin RJ, et al. The effect of preinjury sleep difficulties on neurocognitive impairment and symptoms after sport-related concussion. Am J Sports Med 2015;43:830–838.
57. Levitch CF, McConathey E, Aghvinian M, Himmelstein M, Lipton ML, Zimmerman ME. The impact of sleep on the relationship between soccer heading exposure and neuropsychological function in college-age soccer players. J Int Neuropsychol Soc 2020;26:633–644.
58. McKenna BS, Dickinson DL, Orff HJ, Drummond SP. The effects of one night of sleep deprivation on known-risk and ambiguous-risk decisions. J Sleep Res 2007;16:245–252.
59. Venkatraman V, Chuah YM, Huettel SA, Chee MW. Sleep deprivation elevates expectation of gains and attenuates response to losses following risky decisions. Sleep 2007;30:603–609.
60. Telzer EH, Fuligni AJ, Lieberman MD, Galván A. The effects of poor quality sleep on brain function and risk taking in adolescence. Neuroimage 2013;71:275–283.
61. Short MA, Weber N. Sleep duration and risk-taking in adolescents: a systematic review and meta-analysis. Sleep Med Rev 2018;41:185–196.
62. Sargent C, Lastella M, Romyn G, Versey N, Miller DJ, Roach GD. How well does a commercially available wearable device measure sleep in young athletes?. Chronobiol Int 2018;35:754–758.
63. Schneider KJ, Leddy JJ, Guskiewicz KM, et al. Rest and treatment/rehabilitation following sport-related concussion: a systematic review. Br J Sports Med 2017;51:930–934.
64. Mychasiuk R, Hehar H, Ma I, Candy S, Esser MJ. Reducing the time interval between concussion and voluntary exercise restores motor impairment, short-term memory, and alterations to gene expression. Eur J Neurosci 2016;44:2407–2417.
65. Harmon KG, Clugston JR, Dec K, et al. American Medical Society for Sports Medicine position statement on concussion in sport. Br J Sports Med 2019;53:213–225.
66. Misch MR, Raukar NP. Sports medicine update: concussion. Emerg Med Clin North Am 2020;38:207–222.
67. Niederer D, Engeroff T, Lange K, Vogt L, Banzer W. Return-to-play after concussion: state of knowledge, frequency of use and application barriers of guidelines among decision-makers in rugby. Brain Inj 2018;32:1096–1102.
68. Olympia RP, Ritter JT, Brady J, Bramley H. Return to learning after a concussion and compliance with recommendations for cognitive rest. Clin J Sport Med 2016;26:115–119.
69. Marshall S, Bayley M, McCullagh S, et al. Updated clinical practice guidelines for concussion/mild traumatic brain injury and persistent symptoms. Brain Inj 2015;29:688–700.
70. Lumba-Brown A, Yeates KO, Sarmiento K, et al. Diagnosis and management of mild traumatic brain injury in children: a systematic review. JAMA Pediatr 2018;172:e182847.
71. Lumba-Brown A, Yeates KO, Sarmiento K, et al. Centers for disease control and prevention guideline on the diagnosis and management of mild traumatic brain injury among children. JAMA Pediatr 2018;172:e182847.
72. Walton SR, Kerr ZY, Brett BL, et al. Health-promoting behaviours and concussion history are associated with cognitive function, mood-related symptoms and emotional-behavioural dyscontrol in former NFL players: an NFL-LONG Study. Br J Sports Med 2021;55:683–690.
73. Raikes AC, Dailey NS, Forbeck B, Alkozei A, Killgore WDS. Daily morning blue light therapy for post-mTBI sleep disruption: effects on brain structure and function. Front Neurol 2021;12:625431.
74. Barlow KM, Kirk V, Brooks B, et al. Efficacy of melatonin for sleep disturbance in children with persistent post-concussion symptoms: secondary analysis of a randomized controlled trial. J Neurotrauma 2021;38:950–959.
75. Morawska MM, Büchele F, Moreira CG, Imbach LL, Noain D, Baumann CR. Sleep modulation Alleviates axonal damage and cognitive decline after rodent traumatic brain injury. J Neurosci 2016;36:3422–3429.
76. Hugon J, Hourregue C, Cognat E, et al. Chronic traumatic encephalopathy. Neurochirurgie 2021;67:290–294.

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.

77. Costanza A, Radomska M, Zenga F, et al. Severe suicidality in athletes with chronic traumatic encephalopathy: a case series and overview on putative ethiopathogenetic mechanisms. Int J Environ Res Public Health 2021;18.
78. Cherry JD, Babcock KJ, Goldstein LE. Repetitive head trauma induces chronic traumatic encephalopathy by multiple mechanisms. Semin Neurol 2020;40:430–438.
79. Alosco ML, Culhane J, Mez J. Neuroimaging biomarkers of chronic traumatic encephalopathy: targets for the academic memory disorders clinic. Neurotherapeutics 2021;18:772–791.
80. Ayubcha C, Revheim ME, Newberg A, et al. A critical review of radiotracers in the positron emission tomography imaging of traumatic brain injury: FDG, tau, and amyloid imaging in mild traumatic brain injury and chronic traumatic encephalopathy. Eur J Nucl Med Mol Imaging 2021;48:623–641.
81. Mantyh WG, Spina S, Lee A, et al. Tau positron emission tomographic findings in a former US football player with pathologically confirmed chronic traumatic encephalopathy. JAMA Neurol 2020;77:517–521.
82. Grashow R, Weisskopf MG, Baggish A, et al. Premortem chronic traumatic encephalopathy diagnoses in professional football. Ann Neurol 2020;88:106–112.
83. Quan SF. Are sleep disturbances a risk for chronic traumatic encephalopathy? Only the shadow knows. J Clin Sleep Med : JCSM : official Publ Am Acad Sleep Med 2014;10:241–242.
84. Adams JW, Alosco ML, Mez J, et al. Association of probable REM sleep behavior disorder with pathology and years of contact sports play in chronic traumatic encephalopathy. Acta Neuropathol 2020;140:851–862.

Copyright © by the American Clinical Neurophysiology Society. Unauthorized reproduction of this article is prohibited.